Recommender Systems

- Introduction -

https://youtu.be/7eUxFyo-OVY

Institute for Software Technology
Inffeldgasse 16b/2
A-8010 Graz
Austria

Course Goal

- ☞ Basic introduction to recommendation technologies.

Questions

Which digital camera should I buy?

What is the best holiday destination for me and my family?

Which is the best investment for supporting the education of my children?

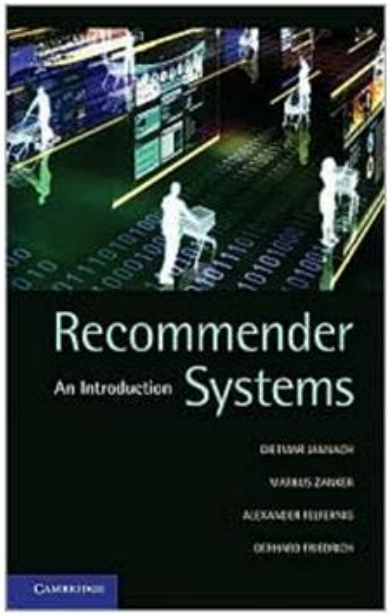
Which movie should I rent?

Which web sites will I find interesting?

Which book should I buy for my next vacation?

Which degree and university are the best for my future?

Overview



Recommender Systems: An Introduction

by Dietmar Jannach, Markus Zanker, Alexander Felfernig, Gerhard Friedrich

AVERAGE CUSTOMER RATING:

(Be the first to review)

Gefällt mir — Registrieren, um sehen zu können, was deinen Freunden gefällt.

FORMAT:
Hardcover

NOOKbook (eBook) - not available

Tell the publisher you want this in NOOKbook format

NEW FROM BN.COM

~~$65.00~~ List Price

$52.00 Online Price

(You Save 20%)

Add to Cart

NEW & USED FROM OUR

New starting at **$56.46** (You S

Used starting at **$51.98** (You S

See All Prices

Table of Contents

Customers who bought this also bought







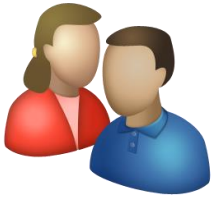




Problem Domain

Recommendation systems (RS) help to match users with items

- Counteract information overload
- Sales assistance (guidance, advisory, persuasion,…)



RS are software agents that elicit the interests and preferences of individual consumers […] and make recommendations accordingly.

They have the potential to support and improve the quality of the decisions consumers make while searching for and selecting products online.

» (Xiao & Benbasat 2007[1])



Different system designs / paradigms

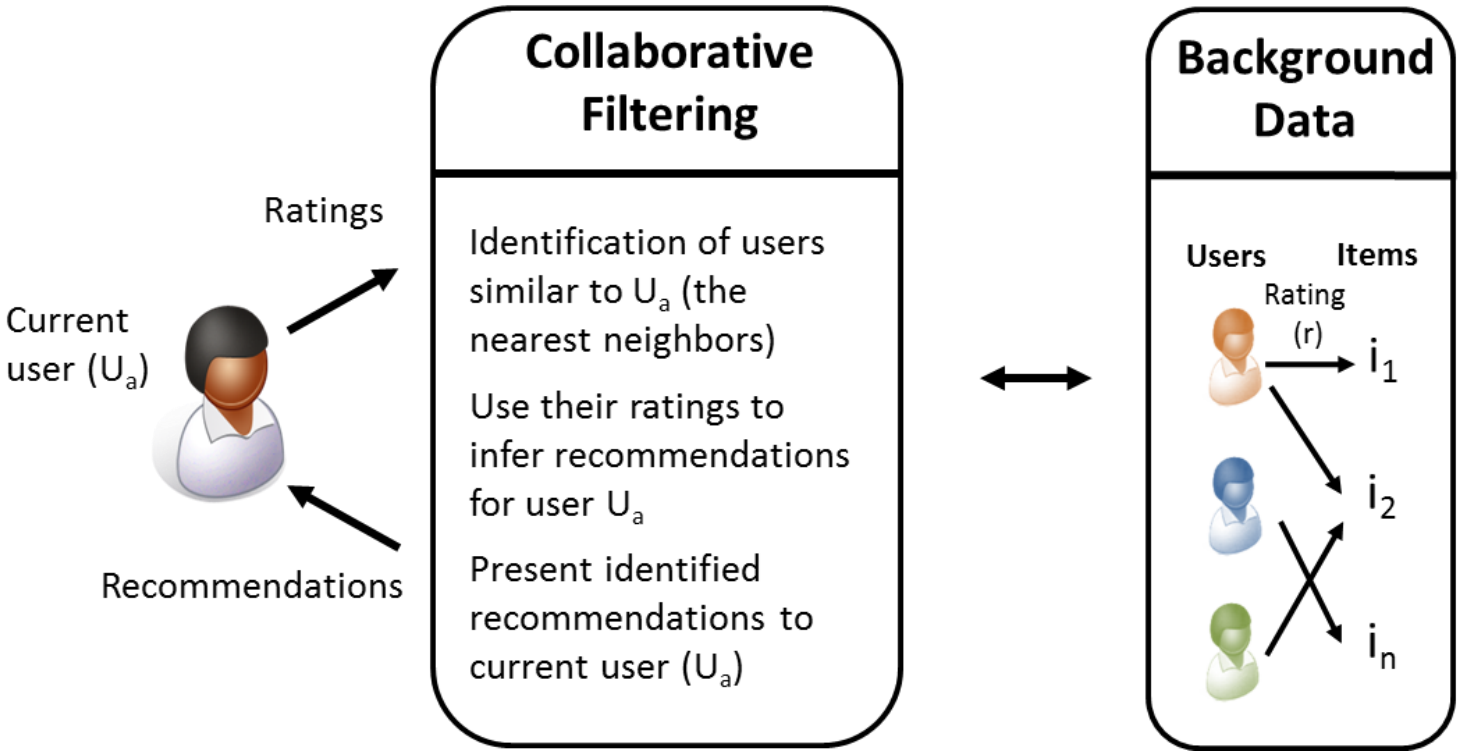
- Based on availability of exploitable data
- Implicit and explicit user feedback
- Domain characteristics



(1) Xiao and Benbasat, *E-commerce product recommendation agents: Use, characteristics, and impact*, MIS Quarterly 31:1, 137–209, 2007.

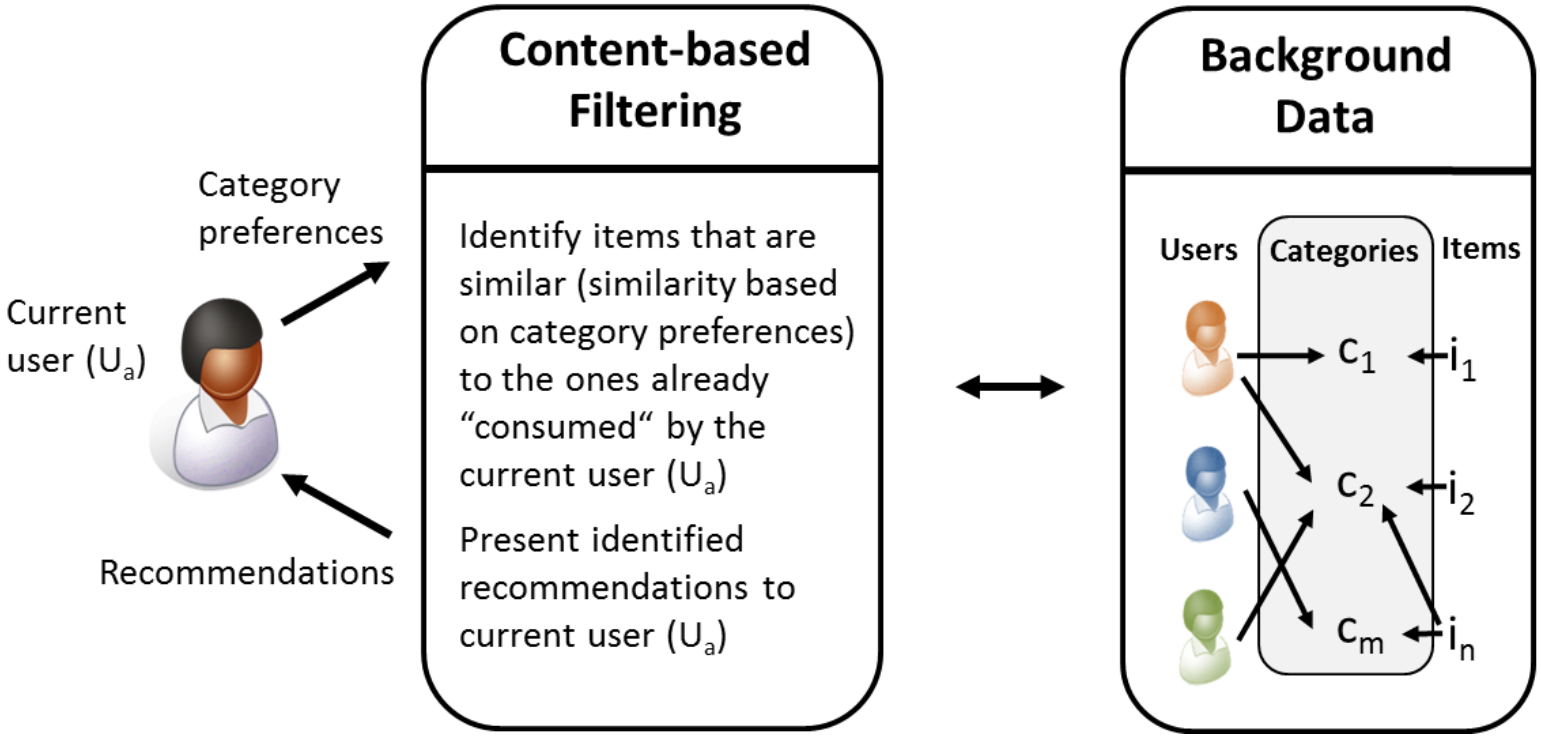
Collaborative Filtering

Collaborative: "Tell me what's popular among my peers"



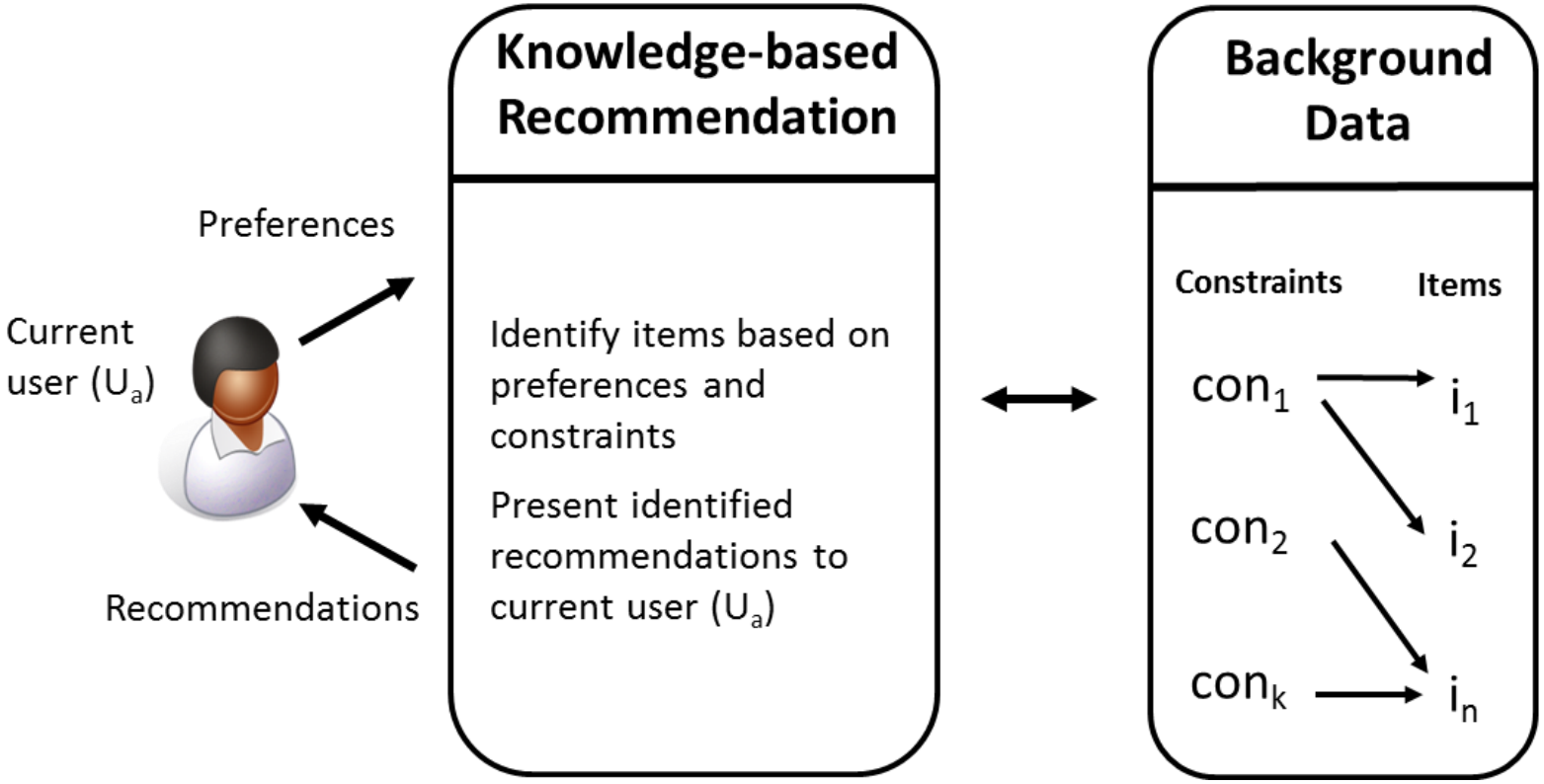
Content-based Recommendation

Content-based: "Show me more of the same what I've liked"



Knowledge-Based Recommendation

Knowledge-based: "Tell me what fits based on my needs"



Further Recommendation Approaches

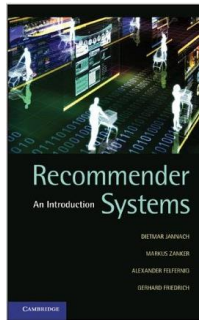
- **Group recommenders**: recommendations not for single persons but for whole groups
- **Critiquing-based** and **case-based recommenders**: also rely on semantic knowledge about items
- **Hybrid recommenders**: combination of different basic paradigm
- **Advanced algorithms**, e.g., matrix factorization, deep learning based recommender systems

Exercise

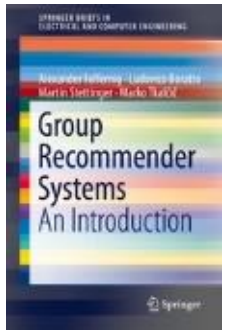
- Propose one product domain for each of the discussed basic recommendation approaches and explain your proposal
  - Collaborative Filtering
  - Content-based Filtering
  - Knowledge-based Recommendation
- Note: no submission („Abgabe“) needed!

Course Material

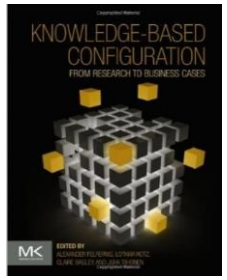
- D. Jannach, M. Zanker, A. Felfernig, and G. Friedrich. Recommender Systems – An Introduction, Cambridge University Press, 2010.



- A. Felfernig, L. Boratto, M. Stettinger, and M. Tkalcic, Group Recommender Systems, Springer, 2018.



- A. Felfernig, L. Hotz, C. Bagley, and J. Tiihonen. Knowledge-based Configuration – From Research to Business Cases, Morgan Kaufmann, 2014.



Thank You!